Is K3,3 a Diophantine graph ?

Problèmes ouverts — A. Dujella · 2026

Problem 1.4 (Dujella, 2026). Do there exist distinct positive integers $a_1, a_2, a_3, b_1, b_2, b_3$ with the property that $a_i b_j + 1$ are perfect squares for each $1 \leqslant i, j \leqslant 3$?

Remark. In the terminology of Batta, Hajdu and Pongrácz (J. London Math. Soc. (2) **111** (2025), Paper No. e70163), the problem asks whether $K_{3,3}$ is a Diophantine graph. Note that $K_{2,t}$ is Diophantine for every t, while K_5 is not Diophantine by the theorem of He-Togbé-Ziegler.